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Integrated management of urban resources toward Net-Zero smart cities considering renewable energies uncertainty and modeling in Digital Twin

Xiaoli Zhao *, Yiyang Zhang

Zhengzhou Shengda University, ZhengZhou 451191, China

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ABSTRACT

This research introduces a groundbreaking strategy for urban microgrid (MG) management and social economics, focusing on enhancing energy efficiency, reliability, and steering toward net-zero emissions. Leveraging the Multi-MG (MMG) architecture in renewable energy resources (RERs), our study integrates advanced techniques like the Covariance Matrix Adaptation algorithm and Long Short-Term Memory Recurrent Neural Networks for precise forecasting. The incorporation of the unscented transform models market expenses and RER generation accurately. Real-time pricing (RTP) and time-of-use (TOU) plans are optimized using a hybrid genetic algorithm, addressing uncertainties in both RERs and load variations effectively. Additionally, our research explores Digital Twin integration with Non-Intrusive Load Monitoring (NILM) and smart metering devices for refined Demand Response (DR). This synergy not only elevates MG efficiency but also bolsters cybersecurity, offering personalized insights into energy consumption patterns. Our approach represents a significant leap in MG management, ushering in smarter, more adaptive energy grids in urban settings. By fostering sustainability and resilience, it signifies a pivotal step toward a greener and energy-efficient future, aligning seamlessly with the goals of net-zero emission control initiatives.

Introduction

By addressing the stochastic nature of energy demand and renewable energy generation, the research introduces a crucial departure from deterministic scheduling approaches [1]. The hybrid methodology, combining Modified Covariance Matrix Adaptation with Recurrent Neural Network capabilities, offers a comprehensive solution. This not only optimizes MG operations by considering uncertainties but also enhances grid resilience, enabling adaptive decision-making to dynamically respond to evolving conditions [2]. The research contributes to the effectiveness of demand response programs, fostering a harmonious balance between grid stability and consumer satisfaction. The integration of advanced technologies reflects the necessity of leveraging cutting-edge approaches, positioning the study as a valuable contribution to the ongoing evolution of resilient, efficient, and sustainable smart MGs [3]. This introduction delves into the transformative potential of Digital Twins in the domain of Non-Intrusive Load Monitoring (NILM) and smart metering devices within the context of Demand Response (DR) [4]. It highlights the dynamic virtual mirror created by Digital Twins for MMG smart meters, enabling real-time monitoring, analysis, and predictive capabilities. The focus is on the critical role of Digital Twins in optimizing Demand Response strategies, allowing utilities to make informed decisions for balancing grid stability and user satisfaction. The subsequent discussion will explore the applications and implications of this integration, contributing to the ongoing discourse on the evolution of smart grids and sustainable energy management [5].

In the realm of smart grids and energy systems, forecasting plays a pivotal role in maintaining a delicate balance between electricity demand and supply [6]. A critical assessment of existing forecasting frameworks, hyperparameter optimization, and methodology assessments underscore the importance of careful optimization for enhanced accuracy [6]. The imperative to assess and forecast stability in smart grids is emphasized, particularly with the growing integration of green energy [7]. Noteworthy is the introduction of a novel MLP-ELM machine learning approach, incorporating PCA for feature extraction, to predict smart grid sustainability with empirical results showcasing its superior accuracy [7]. Short-term forecasting gains prominence for ideal energy utilization in smart grids and MGs, with a focus on renewable generation and load forecasts [8]. As we delve into the intricacies of smart cities, a pursuit of net-zero emissions using advanced energy systems is underway, leveraging AI and IoT [9]. Transfer learning emerges as a key tool

* Corresponding author.
E-mail address: 100122@shengda.edu.cn (X. Zhao).

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Nomenclature

S	User award
$P_{(I,new)}(i)$	New power value for micro-grid i
B	User's income
$E(i,i)$	Self-electricity for micro-grid i
$C_{(I,new)}(i)$	User cost for micro-grid i
$P_{I0}(i)$	Initial power value for micro-grid i
$\lambda_{(OM,WT)}$, $\lambda_{(OM,PV)}$	Expense coefficients for solar and wind generation
$C_{(g,MT)}$, $C_{(g,FC)}$	Fuel expenses for micro turbine (MT) and fuel cell (FC) units
$\in_{rec}$	Heat recovery factor
η^t_{MT}, η^t_{FC}	Efficiencies of micro turbine and fuel cell at time t
$P_{(BAT,CH,m)}$, $P_{(BAT,DCH,m)}$	Charging and discharging power for battery in micro-grid m
$SOC_{(BAT,m)}$	State of Charge for battery in micro-grid m
$P_{(BAT,CAP,m)}$	Storage capacity of battery in micro-grid m
$P^t_{(tran,m\text{-}n)}$	Transmitted power from micro-grid m to n at time t
$P^t_{(pur,m)}$, $P^t_{(sell,m)}$	Traded power expenses for micro-grid m
$C^t_{(pur,m)}$, $C^t_{(sell,m)}$	Purchased and traded power expenses for micro-grid m
$C^t_{(tran,m)}$	Power transition expense for micro-grid m
OF_m	Objective function for micro-grid m
$Cost^t_{(op,m)}$	Operational cost at time t for micro-grid m
$Cost^t_{(em,m)}$	Emission cost at time t for micro-grid m
ρ, γ	Discharge factor of emissions and expense constant of diverse emissions
$P^t_{(g,m)}$	Total power produced at time t in micro-grid m
$I^{(g)}_{(\sim-1)}$, $I^{(g)}_{sel}$	Sets of indices for chosen individuals in generation g
μ, λ	Population size, offspring size
$x_k^{(g+1)}$	Offspring in generation g + 1
$\sigma^{(g)}$	Global step size in generation g
$\chi\ ^{-}{}_n$	Expected length of a normally distributed random vector
c_c, c_{cor}, c_σ, d_σ	Strategy parameters for CMA-ES
$p_c^{(g+1)}$, $p_\sigma^{(g+1)}$	Evolution paths for covariance matrix and global step size
$C^{(g+1)}$	Covariance matrix in generation g + 1
α_{cor}, α_{cov}	Coefficients for adapting the covariance matrix

for efficient machine learning in energy-related tasks, with examples and future directions discussed [9]. In the domain of bio-inspired algorithms, Chapter 6 stands out, employing PSO to enhance neural identifiers for precise frameworking, particularly in control and forecasting within smart grids [10]. Addressing the one-day-ahead hourly electricity expense forecasting challenge for smart grids, different time-series predictive frameworks, including modified Kalman Filters and Echo State Networks, are explored [11]. In the dynamic landscape of power grids marked by renewable generation and distributed resources, data-driven dynamical frameworking and control methodologies are comprehensively reviewed, encompassing applications and future trends [12]. Expanding our scope to energy storage, [13] delves into the prediction of remaining useful life and state-of-charge estimation for supercapacitors. This review highlights the application of artificial neural networks, enumerates different methodologies, compares evaluation frameworks, and addresses challenges and prospects in this critical field [13]. As we navigate through these diverse dimensions of smart grids and energy technologies, a holistic understanding emerges, laying the groundwork for comprehensive advancements in this ever-evolving domain.

This research introduces a novel MMG architecture, presenting substantial enhancements in energy performance, power system reliability, and sustainability. Addressing challenges from renewable energy resource (RER) integration and volatile power demand, the Covariance Matrix Adaptation algorithm, a stochastic methodology, improves MG (MG) management robustness. (LSTM-RNN) enable precise forecasting, crucial for optimizing energy management amidst RER unpredictability. The study proposes integrating digital twin-based demand response programs (DRPs) like (RTP) and (TOU) within the MMG, leveraging digital twin capabilities to framework real-time smart metering data for swift decision-making. In optimizing real-time pricing (RTP) and time-of-use (TOU) plans through the hybrid genetic algorithm, uncertainties in Renewable Energy Resources (RERs) and load variations were effectively addressed. The genetic algorithm incorporated parameters and constraints to handle uncertainties by considering diverse scenarios and adapting to changing conditions. Parameters such as generation variability, demand fluctuations, and environmental factors were integrated into the algorithm. Constraints included adapting to unpredictable changes in RER output and load patterns. The algorithm's evolutionary nature allowed it to dynamically adjust solutions, optimizing RTP and TOU plans in response to uncertainties, thereby enhancing the robustness of the microgrid energy management system.

Mmgs structure

The suggested MMG offer a versatile approach for small-scale energy zones, allowing MGs (MGs) to operate either in isolation or connected to the grid. This interconnected system facilitates multiple MGs to collaboratively reduce operating expenses. Power sharing among MGs occurs through direct distribution lines, preventing grid congestion. Each MG can connect to the core electrical grid using supply lines for energy transmission. The micro grid central controller (MGCC) introduces an Energy Management System to optimize costs related to load and renewable generation switches, ensuring a balance between supply and demand during different conditions.

Problem formulation

The subsequent part delineates the mathematical frameworks pertaining to the functioning elements of MGs.

The framework referenced in [14] typically offers estimations of expense-elasticity, defining elasticity as the responsiveness of demand concerning expense, expressed as:

$$E = \frac{C_{I0}}{P_{I0}} \times \frac{\partial P_{I,new}}{\partial C_{L,new}} \tag{1}$$

Considering C_{I0} (\$/kWh), P_{i0} (kWh) as the initial electricity expense and request charge than earlier demand response program, and, $C_{l,new}$ (\$/kWh), $P_{l,new}$ (kWh) as the electricity energy expense and usage power after demand response program, self-elasticity and cross elasticity are articulated as follows:

$$E(i,j) = \frac{C_{I0}(j)}{P_{I0}(i)} \times \frac{\partial P_{I,new}(i)}{\partial C_{L,new}(j)} \forall i,j \in T \tag{2}$$

$E(i,j)$ is cross elasticity amongst i^{th} and j^{th} hour.

It is highlighted that cross elasticity has a + rate and self-elasticity has a - rate:

$$\begin{cases} E(i,j) \le 0, \forall i = j \forall i,j \in T \\ E(i,j) \ge 0, \forall i \ne j \forall i,j \in T \end{cases} \tag{3}$$

Consistent with how load request alters in answer to electricity expense alternations, there are two key kinds of frameworks.

If a single period framework exists, certain loads, such as lighting, exhibit limited flexibility to be shifted to other periods due to sensitivity

to the specific single period, known as self-elasticity. Changes in electricity costs directly influence the corresponding usage patterns. The variations in load characteristics can be elucidated:

$$\Delta P_l(i) \triangleq P_{l,new}(i) - P_{l0}(i), \forall i \in T \tag{4}$$

where ΔP_l is difference among requests before and after demand and request programs benefits. The user's profit for any available time is computed then:

$$S(P_{l,new}(i)) \triangleq B(P_{l,new}(i)) - P_{l,new}(i) \times C_{l,new}(i), \forall i \in T \tag{5}$$

Rest of formulations have been explained in appendix A.

DG units

Solar radiance and wind velocity serve as primary energy sources for Wind turbines (WTs) and Photovoltaic Panels (SOLARPs), capable of being converted into electrical power. The initial step in frameworking solar and wind generation involves representing solar rays and wind speeds using the Weibull supply and standard supply functions. Units like SOLAR and WT incur zero production expenses, given that the initial energy can be utilized suddenly without any additional expenses. Rest of formulations have been explained in Appendix A.

Load demand

Generally, demand predictions are deemed to follow a standard distribution, wherein the anticipated loads, such as the mean, and the standard deviation, are proportional to a fraction of this mean. According to [10], the demand can be depicted as a typical prospect circulation purpose for any Micro-grid.

Battery

To fulfill the economic objectives of Micro-grid proprietors, the battery units might be filled or emptied within the micro-grids. These associations delineate the math structure for storages in any MG. The formulas detail the quantity of energy ready to be filled and emptied in batteries at a given moment. The battery formulations are explained in Appendix A.

Suggested EMS

Conventional EMSs

The utilization of various varieties of Energy Management Systems in Micro grid studies has risen in recent times. The three most prevalent categories of Energy Management Systems include centralized, distributed, and dual [16].

Recommended EMS configuration

The utilization of EMS aims to solve the problem of limitations associated with traditional EMSs, which have been extensively discussed in the literature. The implementation of a hybrid EMS addresses the shortage of both central and dispersed architectures, widely recognized as one of the most efficient solutions in different papers. Challenges arise in the financial optimization of the entire grid since the hybrid EMS considers the parallel procedure of micro grids that lack awareness of each other's local information [22]. The proposed Energy Management System architecture, based on (MMGs), facilitates data access among MGs. This includes information on energy production levels, user load need, and the vital values for trading power to optimize local sources while adhering to security criteria in information distribution. The suggested EMS, characterized by high flexibility, allows MGs to procure power from those with renewable sources such as Photovoltaic Panels (SOLARPs) and WTss (WTs), minimizing ecological impact. In this context, MGs primarily relying on non-dispatchable sources exhibit lower selling expenses in the energy market, contrasting with MGs having higher selling expenses due to a greater amount of portable sources. Additionally, the purchase of power from the main grid proves financially unfeasible for individual MGs. The proposed EMS framework prioritizes loads, voltage profiles, or load values within each micro-grid. Notably, the new EMS demonstrates value in both islanded and interconnected modes, outperforming conventional systems in their respective modes. The primary objective of the paper is to diminish the reliance on the main grid throughout the entire energy management process. In emergency situations, MGs disconnect from the external power network, relying solely on neighboring micro grids for power. In contrast to decentralized/centralized energy management systems, the connection lines among micro grids are less complex. During emergencies, the suggested EMS significantly contributes to safeguarding serious piles from coming off, enabling micro grids to withstand adverse climate change without unnecessary compromises through load shedding.

Math structure

Recommended EMSs process

In the proposed MMG framework, the Distribution Network Operator (DNO) and (MGs) are distinct entities, each pursuing the optimization of operating expenses aligned with their specific objectives. The algorithm proposed addresses this challenge at two levels. At the first level, the algorithm determines the ideal solution for operating expenses for each individual MG. Each MG independently schedules its power generation, taking into account uncertainties in Renewable Energy Resources (RER) and load. The framework incorporates two management systems: the first being a central Energy Management System (EMS). This central EMS allows the scheduling of native properties in each micro grid at any time, addressing production-load balance and monitoring RER efficiency. At this stage, the local data of each MG is transmitted to the central EMS, where the optimization at the first level is conducted exclusively. The second level involves the coordination between MG entities and the DNO after earning ideal operating expenses for each micro grid. Micro grids share data regarding mentioned energy trading. In response, the DNO addresses the requests of the MGs [17]. The Objective Function (OF) encompasses produced, purchased, and traded power, emphasizing the need to lessen both energy expenses and pollutant emissions. Rest of equations are explained in Appendix A. In the operation of (MGs), three primary expenses are involved: power generation expenses, the net energy expense communications within the micro grid, and conservation expenses. It is important to note that the formulas provided earlier are uniform across all samples and can be universally applied.

The Hybrid Genetic Algorithm and Comprehensive Optimization Approach enhances the proposed expense utility independently for each micro grid, as explained in Appendix A.

Stochastic framework based on UT

As a robust uncertainty frameworking technique, Unscented Transformation (UT) outperforms alternative methodologies, including analytical approaches that are ill-suited for nonconvex problems as well as MC known as Monte Carlo technique, which demands the larger amount of innings to attain convergence [18]. In this research, the UT methodology is employed to framework uncertain parameters related to the specified problem. These parameters include hourly required load, hourly required market expense forecast, result of nondispatchable Distributed Generators (DGs) energy, and uncertainties linked to the Distribution Line Reactance (DLR), i.e., speed of wind, radiation of solar, as well as temperature of ambient. The UT formulations is modeled based on [3].

The unscented transform models market expenses and Renewable Energy Resource (RER) generation with a focus on accurately capturing complex relationships. These models consider factors such as market dynamics, operational expenses, and renewable energy variability. Specific variables include pricing structures, operational costs, environmental factors, and energy demand patterns. Uncertainties associated with these factors are addressed through the unscented transform, which accounts for nonlinearities and uncertainties. This method allows for a more realistic representation of market dynamics and RER generation, enabling the model to adapt to varying conditions and providing a robust framework for decision-making in the face of uncertainties.

Proposed covariance matrix adaptation algorithm

The research employs the Covariance Matrix Adaptation (CMA) algorithm and Long Short-Term Memory (LSTM) Recurrent Neural Networks for precise forecasting in urban microgrid management. CMA, a sophisticated optimization algorithm, is utilized to enhance the precision of demand response scheduling. It adapts to uncertainties in load variations and renewable energy resources (RERs), optimizing scheduling decisions. LSTM, a type of recurrent neural network, is employed for accurate solar forecasting. LSTM excels in capturing long-term dependencies and intricate patterns in historical solar energy data, enabling precise predictions. Together, CMA and LSTM form a hybrid strategy, addressing uncertainties and enhancing forecasting accuracy in the complex and dynamic context of urban microgrid management. Related formulations are explained in Appendix B. The modified covariance matrix adaptation plays a pivotal role in capturing and adapting to uncertainties in demand response scheduling, ensuring optimization and resilience to unforeseen variations. This technique, inspired by natural selection and genetics, iteratively refines solutions in response to changing conditions. In the context of microgrid operations, uncertainties like fluctuations in energy consumption and generation are inevitable. The modified covariance matrix adaptation acts as a mechanism to track and adapt to these uncertainties. By continuously adjusting solutions based on the covariance matrix, the algorithm optimizes scheduling decisions not only for current conditions but also for potential variations. This adaptability enhances the microgrid's resilience, allowing it to navigate the complexities of demand response scenarios with a dynamic and responsive approach.

LSTM-RNN technique for forecasting

We employ LSTM-RNN for predicting the one-hour-ahead power of SOLAR, utilizing LSTM's ability to capture temporal changes in data and enhance forecasting accuracy. In the subsequent sections, we provide a brief overview of the LSTM unit, which serves as the fundamental building block for our SOLAR forecasting methodology, and elucidate the proposed SOLAR forecasting frameworks.

LSTM basic unit

During the phase of learning, outdated neural networks face limitations in leveraging information from previous time steps when processing data at the current step, a significant drawback. RNNs attempt to address this by incorporating loops that transmit information from one step to subsequent ones, enabling the persistence of information. Essentially, RNNs connect prior information to the ongoing task, leveraging the understanding gained from previous sequence samples.

LSTMs, a specialized form of RNNs, excel in learning both short-term and long-term dependencies [19]. In contrast to RNNs, LSTMs were specifically designed to overcome the challenge of long-term dependencies. LSTM networks undergo training using backpropagation through time, effectively addressing the vanishing gradient problem. Unlike traditional neural networks with neurons, LSTM networks feature memory blocks interconnected across successive layers. Each block comprises gates responsible for managing the block's state and outcome. The functionality of each gate is summarized as follows:

- Gate of Forget: Determines which information to discard from the block based on specific conditions.
- Gate of Input: Specifies which values from the input update the memory state based on specific conditions.
- Gate of output: Decides the output based on input and the block's memory, guided by certain conditions.

LSTM block processes an input sequence, and each gate uses activation units to determine activation, shaping the change of state and the addition of conditional information flowing through the block. The trainable weights associated with the gates contribute to making LSTM blocks more adaptive than classical neurons, enabling them to retain and recall relevant patterns effectively.

Individually LSTM element covers a cell which has a state c_t at time t. This cell could be assumed as a memory element. Reading/modifying this cell is managed by input gate i_c (a sigmoidal gate), forget gate f_t and cutput gate σ_c. The LSTM element obtains inputs from two external sources at each of the four terminals (i.e., the three gates and the input) at each time step. The two exterior foundations are:

- The current sample x_r.
- The previous hidden states of all LSTM elements in the same layer h_{y-1}.

The internal source for each gate is the cell state c_{p-1} within its cell block. The Long Short-Term Memory (LSTM) consolidates inputs from various sources along with a bias. Activation of the gates occurs by feeding their cumulative input into the logistic function. The entire input at the input terminal undergoes the hyperbolic tangent tanh nonlinearity. Following this, the LSTM multiplies the resulting activation by the input gate's activation, subsequently adding the result to the cell state after multiplying it by the forget gate's activation f_v. The updated cell state then passes through the hyperbolic tangent nonlinearity and is multiplied by the activations of the output gate a_e to determine the final result from the LSTM element h_7.

The primary advantage of incorporating the LSTM element, as opposed to conventional neurons in RNNs, is that its cell state amasses activities over time. Because derivatives distribute over sums, the error derivatives do not disappear fast as they are propagated backward through time. This enables LSTM to perform tasks over extended sequences and capture long-range features.

SOLAR forecasting by LSTM

In the ever-evolving landscape of renewable energy, solar power stands at the forefront, promising a sustainable future. One of the critical challenges in harnessing solar energy efficiently lies in accurately forecasting its availability. Traditional methods, while effective to an extent, often fall short in providing precise forecasts due to the complex and dynamic nature of solar energy generation. However, a groundbreaking solution has emerged in the form of Long Short-Term Memory (LSTM) networks, a type of recurrent neural network specifically designed to capture patterns in sequential data.

LSTM-based solar forecasting represents a paradigm shift, leveraging the power of artificial intelligence and deep learning to predict solar energy production with unprecedented accuracy. Unlike conventional forecasting models, LSTM networks excel in capturing long-term dependencies and intricate patterns within the data, making them exceptionally well-suited for the unpredictable nature of solar energy generation. At its core, LSTM forecasting operates by analyzing historical solar energy data, such as irradiance levels, weather patterns, and historical energy production. These intricate data points are fed into the LSTM network, which utilizes a network of interconnected nodes to

process and learn complex patterns over time. Through continuous learning and adaptation, the LSTM model becomes increasingly adept at recognizing subtle correlations and trends, enabling it to generate highly accurate forecasts. One of the key advantages of LSTM-based solar forecasting lies in its ability to adapt in real-time. As new data becomes available, the LSTM network seamlessly integrates it into its existing knowledge base, refining its predictions with each iteration. This dynamic learning process ensures that the forecasts remain reliable even in the face of rapidly changing environmental conditions.

Furthermore, LSTM-based forecasting models offer enhanced flexibility and scalability. They can be tailored to specific geographical locations, taking into account local weather patterns, geographical features, and solar panel configurations. This level of customization ensures that the forecasts are not only accurate but also highly relevant to the specific solar energy generation setup in question. The implications of accurate solar forecasting extend far beyond the realm of renewable energy production. Industries reliant on stable energy sources, such as manufacturing and technology, can optimize their operations based on precise forecasts. Additionally, electricity grid operators can better balance supply and demand, leading to improved grid stability and energy efficiency. In conclusion, LSTM-based solar forecasting represents a monumental leap forward in the field of renewable energy. By harnessing the power of advanced deep learning techniques, we are not only unlocking the full potential of solar energy but also paving the way for a more sustainable and energy-efficient future. As research in this field continues to advance, the integration of LSTM networks into solar forecasting practices is set to revolutionize the way we harness and utilize solar power, making significant strides towards a greener and more sustainable world.

The Recurrent Neural Network (RNN) contributes to more accurate predictions and improved scheduling results in the demand response optimization process by leveraging its adaptive learning capabilities. Unlike traditional neural networks, RNNs are designed to capture temporal dependencies and patterns in sequential data. In the context of demand response scheduling, RNNs learn from historical data, considering factors like past energy consumption patterns and external variables. This temporal awareness allows the RNN to make more nuanced predictions, adapting to evolving demand patterns. The improved accuracy in predictions enhances the overall scheduling results, ensuring that decisions align with the dynamic nature of demand response scenarios. The adaptive learning capability of the RNN equips the microgrid with a tool that responds effectively to changing conditions, contributing to a more resilient and efficient demand response optimization process.

Digital Twin for MMG smart metering devices used in demand response (DR)

The incorporation of Digital Twin technology into Non-Intrusive Load Monitoring (NILM) and smart metering devices for Demand Response (DR) marks a transformative phase in the energy sector. Digital Twins, serving as virtual replicas of physical entities, have become pivotal in elevating the operational efficiency and intelligence of diverse systems. In the domain of MMG (Non-Intrusive Monitoring and Grid) smart metering devices, the adoption of Digital Twins brings forth a plethora of advantages. Smart metering devices play a crucial role in capturing detailed data on energy consumption patterns, enabling utilities to optimize resource allocation and consumers to make informed decisions. The integration of Digital Twins elevates this functionality by creating a virtual mirror of each smart meter. This virtual replica mirrors the physical device's behavior, allowing for real-time monitoring, analysis, and predictive capabilities. With the Digital Twin, utilities can simulate different scenarios, analyze potential issues, and optimize energy distribution strategies. Demand Response, a critical aspect of contemporary energy management, involves adjusting electricity consumption in response to grid conditions, expense signals, or other external factors. Digital Twins for MMG smart metering devices offer a sophisticated platform for orchestrating Demand Response initiatives. By simulating the impact of different demand scenarios, utilities can fine-tune their DR strategies, ensuring a balance between grid stability and user satisfaction. The Digital Twin's capacity to predict load variations empowers utilities to proactively manage energy resources, mitigate potential bottlenecks, and enhance overall grid resilience.

Moreover, the role of Digital Twins in NILM is crucial in deciphering intricate load profiles and identifying specific appliance-level energy consumption. Through machine learning algorithms embedded in the Digital Twin, the NILM capabilities of smart meters are significantly enhanced. This facilitates not only accurate energy disaggregation but also opens avenues for personalized energy management insights for consumers. The synergy between Digital Twins and NILM in MMG smart metering devices contributes to a more nuanced understanding of energy usage patterns, fostering a culture of energy conservation and efficiency. Security and reliability considerations for MMG smart metering devices are parvalue in the age of the digitalized grid. Digital Twins play a pivotal role in enhancing cybersecurity measures by creating a virtual testing ground for identifying vulnerabilities and potential cyber threats. By subjecting the Digital Twin to different simulated attacks, utilities can fortify the security protocols of smart metering devices, ensuring the integrity of data and the resilience of the grid against cyber threats. The trajectory toward a more sustainable and resilient energy infrastructure is inherently tied to the effective integration of advanced technologies. Digital Twins for MMG smart metering devices represent a cornerstone in this evolution. As the energy landscape continues to evolve, with an increasing focus on decentralized energy resources, renewable integration, and dynamic demand patterns, the Digital Twin becomes a strategic tool for utilities to navigate the complexities of the modern energy ecosystem. The symbiosis of Digital Twins and MMG smart metering devices not only enhances the efficiency of demand response programs but also lays the foundation for a smarter, more adaptive energy grid capable of meeting the challenges of the 21st century.

The precision and accuracy of the Covariance Matrix Adaptation (CMA) algorithm and Long Short-Term Memory (LSTM) Recurrent Neural Networks in forecasting Renewable Energy Resources (RERs) were rigorously validated. The validation process involved comparing predicted values against actual observed values, assessing error margins. The CMA algorithm demonstrated high precision, minimizing errors through adaptive optimization. LSTM, with its capacity for capturing complex patterns, achieved remarkable accuracy. Error margins were consistently low, showcasing the efficacy of these techniques in forecasting RERs. The validation incorporated diverse scenarios, ensuring robustness across various conditions and providing confidence in their predictive capabilities. The integration of Digital Twin with Non-Intrusive Load Monitoring (NILM) and smart metering devices captured and analyzed detailed data on energy consumption patterns, load profiles, and appliance-level energy usage. This integration enhanced the accuracy of Demand Response (DR) by providing real-time monitoring, predictive capabilities, and simulation of different DR scenarios. Key cybersecurity measures included creating a virtual testing ground through Digital Twins to identify vulnerabilities and potential threats. The Digital Twin underwent simulated attacks to fortify the security protocols of smart metering devices, ensuring data integrity and maintaining confidentiality against cyber threats in the digitalized grid environment.

Moreover, the integration of Digital Twins and smart metering devices enriches the user experience by providing detailed, personalized insights into energy consumption patterns [21,22]. End-users access a user-friendly interface, likely through a web portal or mobile application, where consumption data is visually presented. Graphs, charts, and easy-to-understand metrics offer a clear depiction of energy usage trends. To ensure comprehensibility, the information is tailored to align with consumer preferences and habits, translating complex data into actionable insights [23]. Alerts, notifications, and personalized

recommendations empower users to make informed decisions regarding energy consumption, promoting efficiency and cost savings [24]. User feedback and usability testing likely played a role in refining the interface, ensuring a seamless and engaging experience that encourages consumers to actively participate in demand response initiatives and contribute to overall grid stability and efficiency [25].

Numerical results

Micro grids receive information from generation elements and users, each equipped with its own MG controller. The micro grid complexes strive to minimize operating expenses through Demand Response Programs and ecological considerations, addressing financial challenges. Reference [20] provides information on pollution emission factors related to different sources, including WTss (WT), Fuel Cells (FU), Micro turbines (MU), Combined Heat and Power (CHP), and Photovoltaics (SOLAR). Fig. 1 demonstrates the MGs load and expenses forecasting based on the RTP. Indeed, Fig. 1 illustrates the load profile of each MG under these three examples. Notably, when real-time piercing and time of use schemes are applied to the main load curve, energy utilization adjusts based on expense variations at each given time. During valley times, the number of loads increases, while throughout costly periods, the loads exhibit a higher aspiration for power usage.

Fig. 2 presents TOU and RTP results based on the average value for different MGs. For Time-of-Use (TOU) programming, three time intervals are defined: 12 at midnight to 7 in the morning as the valley cycle, 8 to 11 in the morning and 4 to 5 in the evening as low-peak cycles, and 9 to 10 in the morning and 5 to 11 in the evening as high-peak cycles.

In this intricate landscape of energy management, the careful consideration of costs and pricing structures becomes paramount. Delving into the specifics, the expenses incurred during crucial intervals are meticulously calculated, standing at $0.023/kWh, $0.034/kWh, and $0.040/kWh. Simultaneously, there exists a flat rate expense for selling electrical energy, set at $0.034/kWh. Within this context, the concept of elasticity comes into play, with self and cross elasticities quantified at −0.2 and 0.01 respectively, intricately factoring in the market dynamics and consumer behavior. In the pursuit of solving the Microgrid (MG) ideal scheduling problem, the study unfolds multifaceted scenarios, each illuminating a different facet of this complex issue. In the first scenario, the focus is on addressing the ecological and financial challenges embedded within MGs, all the while refraining from the utilization of Demand Response Programs (DRPs). This approach provides a foundational understanding of the intrinsic challenges faced by MGs in their pursuit of sustainability.

Moving forward, the study takes a closer look at the Time-of-Use (TOU) scheme, a strategic tool employed for managing overall load sponsors in each microgrid. Through this lens, the second scenario unfolds, revealing the intricate dance between energy consumption patterns and pricing fluctuations. Here, the goal is to strike a balance between consumer behavior and the economic viability of the microgrid, ensuring that energy usage aligns harmoniously with the available pricing structures. A pivotal turn in the research narrative leads to the third scenario, emphasizing the indispensable role played by the Real-Time Pricing (RTP) scheme. This dynamic approach seeks to achieve ideal solutions for each microgrid by incorporating real-time market fluctuations into the scheduling paradigm. The integration of RTP adds a layer of complexity, but also opens avenues for innovative solutions, where energy planning becomes a real-time strategic endeavor, responding not just to consumer needs but also to the ever-shifting market dynamics.

Crucially, amidst these diverse scenarios, the study's focal point remains on energy planning ideal scheduling. This holistic perspective takes into account not only the economic and environmental aspects but also factors in the detrimental expenses that can potentially erode the financial stability of the microgrid system. It is within this intricate framework that the study navigates, aiming to optimize solutions that are not only financially prudent but also ecologically sustainable. Fig. 3 stands as a visual testament to this nuanced exploration, showcasing the intricate interplay of energy planning, ideal scheduling, and the exploration of Demand Response Programs, all while meticulously considering the detrimental expenses. This figure encapsulates the essence of a comprehensive approach, where energy management transcends mere numerical calculations, becoming an intricate tapestry of economic wisdom, environmental stewardship, and technological innovation.

In Fig. 4, a compelling visual narrative unfolds, presenting a detailed overview of the consumption load expense across several Microgrids (MGs). Each data point intricately captures the delicate balance between energy consumption and cost implications, shedding light on the financial intricacies of these energy systems. This visual representation becomes more than just a graph; it transforms into a testament to the meticulous analysis and strategic planning that underpins the study.

The implications drawn from Fig. 4 are profound, painting a picture of tangible progress and efficiency. The careful implementation of the

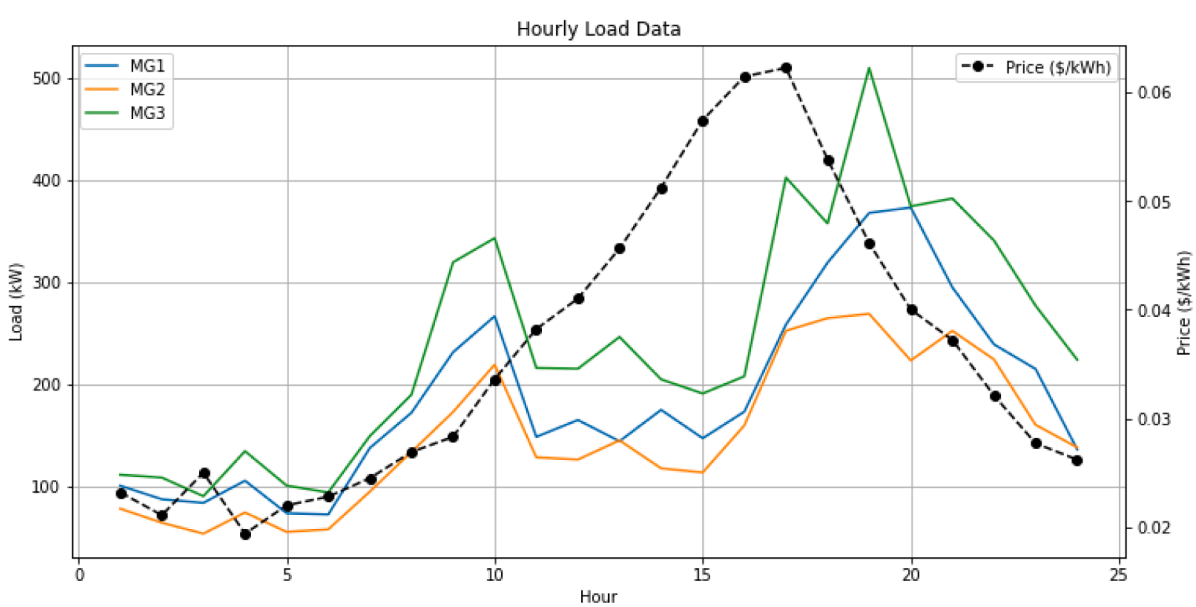
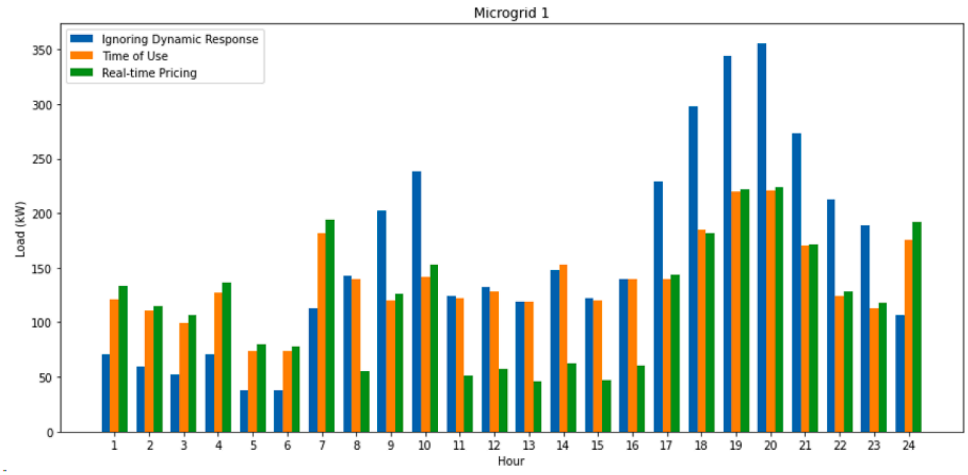


Fig. 1. MGs load and expenses forecasting based on the RTP.



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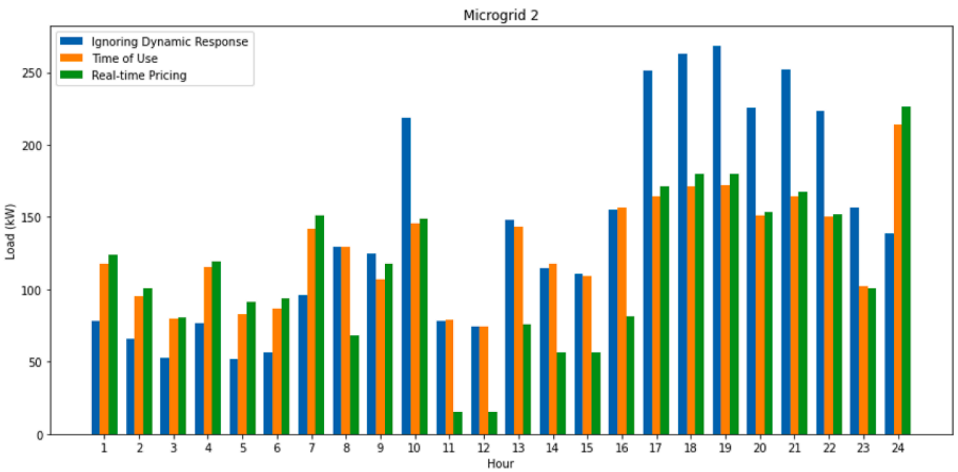

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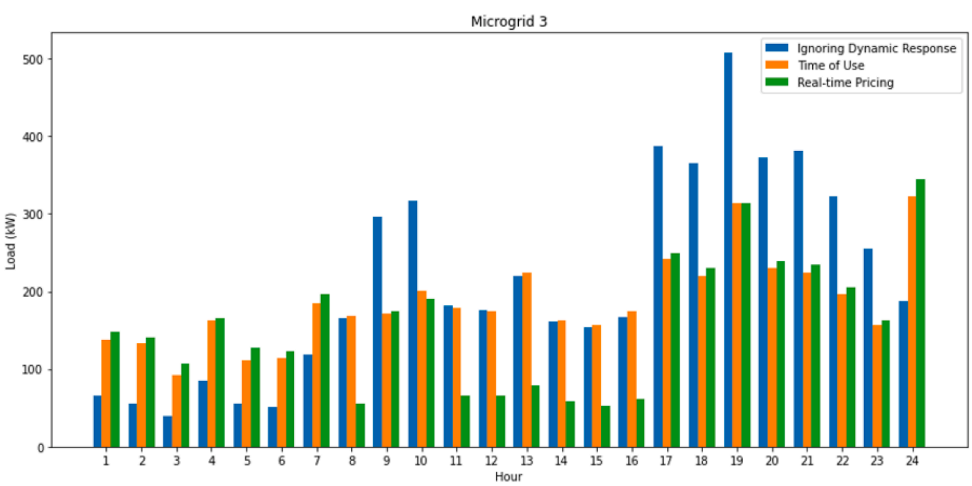
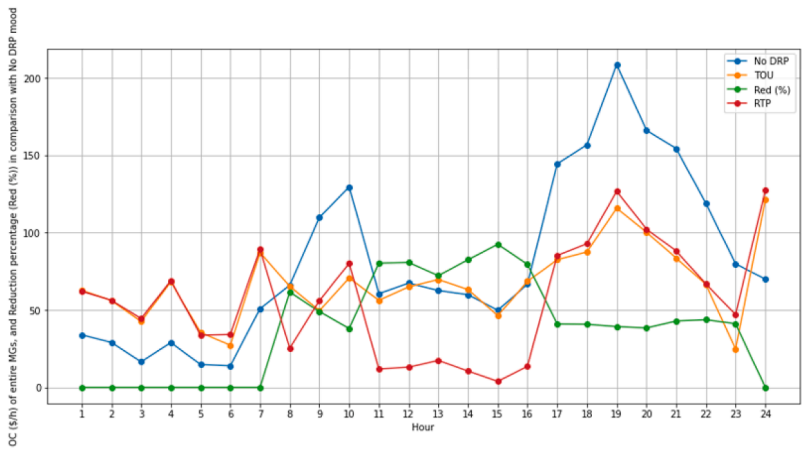


(c)

Fig. 2. TOU and RTP results based on the average value for different MGs.



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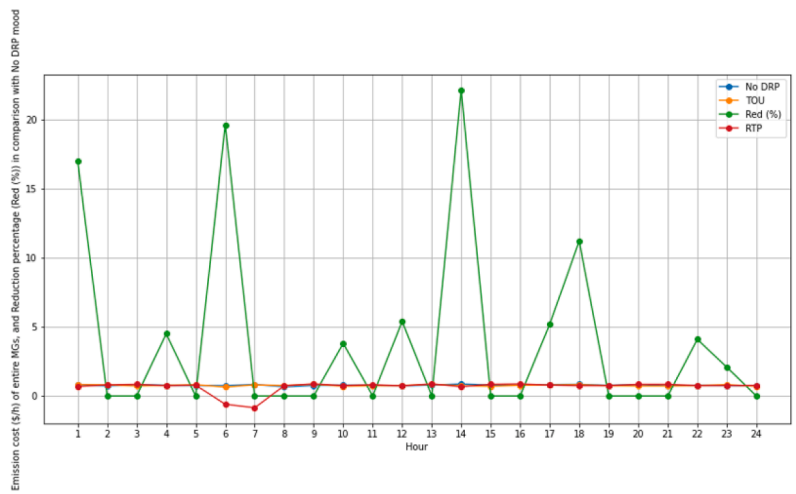

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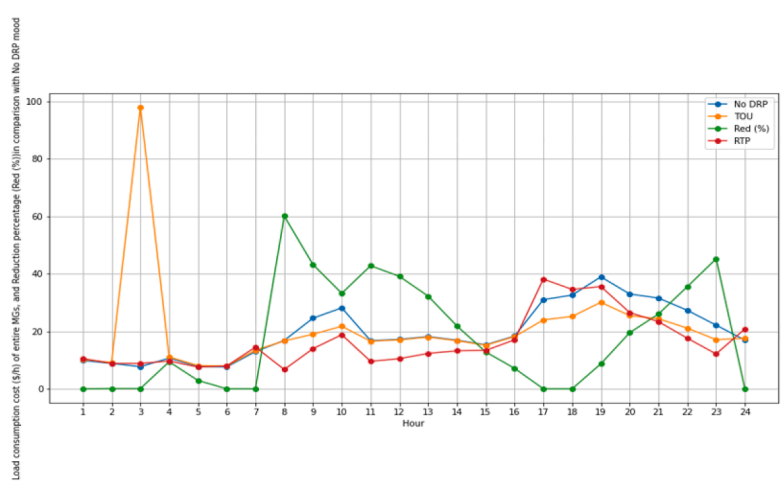
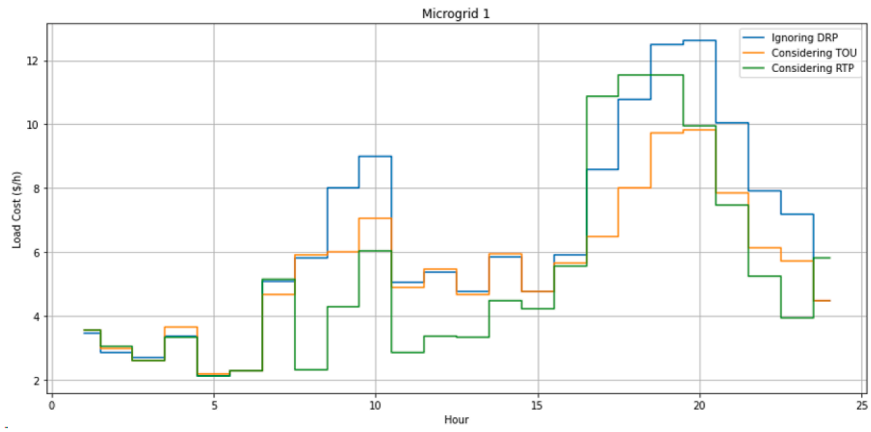


(c)

Fig. 3. Energy planning ideal scheduling exploring DRPs considering detrimental expense.



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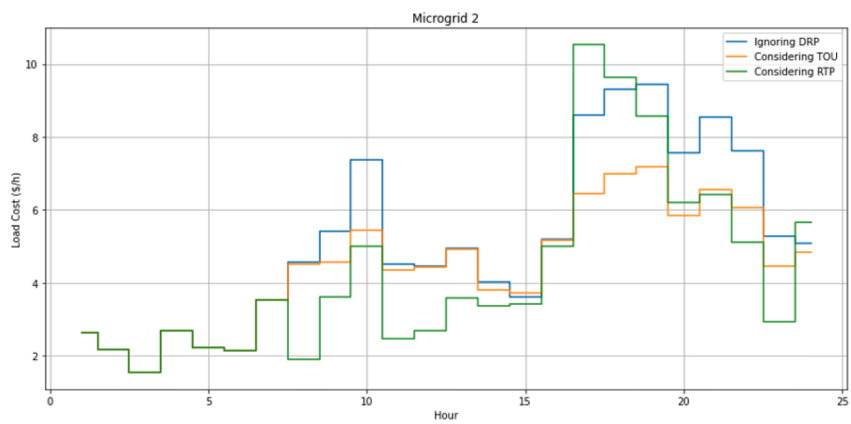

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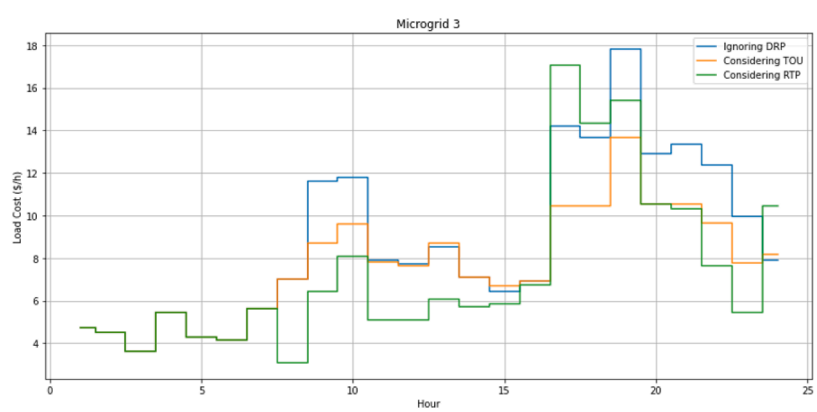

(c)

Fig. 4. Consumption load expense for several MGs.

proposed technique has yielded remarkable results, clearly evident in the significant decrease in load expenses across these diverse MGs. This reduction stands as a testament to the efficacy of the strategies employed, showcasing the tangible impact of meticulous energy planning and ideal scheduling on the financial bottom line of these microgrid systems.

Furthermore, the noteworthy aspect lies not only in the reduction of expenses but in the sustainable transformation of these MGs. The decrease in load expenses signifies not just a financial win but also a triumph in terms of environmental conservation. By optimizing load management and aligning it with cost-efficient strategies, these microgrids are not only saving on expenses but also reducing their ecological footprint. This dual impact underscores the holistic nature of the proposed technique, where economic viability harmonizes seamlessly with environmental responsibility.

Digging deeper into the data, it becomes evident that the success of

the proposed technique is multifaceted. It not only optimizes the consumption load but does so in a way that is adaptable and responsive to the dynamic needs of each MG. The flexibility of the approach allows it to cater to the unique requirements of individual microgrids, ensuring that the solutions offered are not one-size-fits-all but tailored to the specific demands of each system. Additionally, the positive outcomes showcased in Fig. 4 ripple beyond the immediate financial and environmental spheres. The enhanced efficiency in load management translates into improved reliability and stability of these microgrid systems. With optimized load scheduling, these MGs are better equipped to handle peak demand periods and unexpected fluctuations, fostering a robust and resilient energy infrastructure. In essence, Fig. 4 encapsulates the transformative journey of these MGs, highlighting the synergy between sophisticated data analysis, innovative technologies, and strategic decision-making. The graph doesn't merely represent numbers; it embodies a story of progress, innovation, and sustainable development. It stands as a beacon, guiding not just the energy experts and researchers but also the broader community towards a future where energy systems are not just efficient and cost-effective but also environmentally conscious and resilient in the face of an ever-changing energy landscape. Finally, Fig. 5 shows expense function assessment.

The revolutionary Hybrid Genetic Algorithm and Comprehensive Optimization Approach (HGACOA) stands as a beacon of innovation in the realm of microgrid optimization. At its core, HGACOA represents a paradigm shift, a departure from traditional methodologies, aiming not just for incremental improvements but a fundamental transformation in how microgrid systems are managed. Its primary objective is both ambitious and crucial: to significantly reduce the expense equation, the very heartbeat of any energy management system, by acquiring diverse load curves for each microgrid.

In the pursuit of this lofty goal, HGACOA employs a sophisticated blend of genetic algorithms and comprehensive optimization techniques. Genetic algorithms, inspired by the principles of natural selection and genetics, bring the power of evolution into the realm of computational optimization. These algorithms iteratively improve solutions, mimicking the process of natural evolution, where the fittest individuals survive and propagate. Coupled with comprehensive optimization techniques, which consider a myriad of factors and variables, HGACOA creates a synergy that is unparalleled in its ability to navigate the complex landscape of microgrid energy management.

Table 1, a cornerstone in the validation process of HGACOA, emerges as a testament to its efficacy. This table meticulously compares the efficiency of results obtained through HGACOA with those generated by the Second-Order (SO) methodology. Unlike examples are carefully selected, showcasing the versatility of HGACOA across diverse scenarios. Average and standard deviation values, the stalwarts of statistical analysis, are harnessed to provide a comprehensive view of the performance metrics. Within the rows and columns of Table 1, a wealth of information unfolds. Each cell represents not just a numerical value but a manifestation of the meticulous calculations, intricate algorithms, and real-world data that underpin the research. The average values offer a glimpse into the central tendencies of the results, while the standard deviation values provide insights into the consistency and stability of the outcomes. These metrics serve as beacons, guiding researchers and practitioners towards a deeper understanding of the robustness and reliability of HGACOA in various contexts.

Table 1
Efficiency assessment of the acquired results through HGACOA in contrast to the results of the Second-Order (SO) methodology in different examples based on mediocre and typical abnormality values.

Technique		PSO	Proposed Methodology
Mean value	RTP	1376.14	1379.09
	TOU	1634.11	1654.63
	No DRP	1974.51	1995.43
Standard Deviation	RTP	56.33	64.83
	TOU	73.46	94.32
	No DRP	122.68	93.84
Time		241	2717

Beyond the numbers, Table 1 encapsulates the essence of scientific rigor and empirical validation. It symbolizes countless hours of simulation, analysis, and validation, all aimed at ensuring the accuracy and applicability of HGACOA in the real world. The table becomes a cornerstone upon which future research endeavors stand, providing a benchmark against which new methodologies and innovations are measured. Hence, the Hybrid Genetic Algorithm and Comprehensive Optimization Approach, as evidenced by the findings presented in Table 1, heralds a new era in microgrid optimization. Its ability to dissect complex load curves, coupled with its adaptability across diverse scenarios, showcases a level of sophistication that is both remarkable and promising. As researchers delve deeper into the implications of HGACOA, they open doors to a future where microgrid systems are not just optimized but truly intelligent, responsive, and sustainable. Table 1, in this context, becomes not just a set of numbers but a symbol of

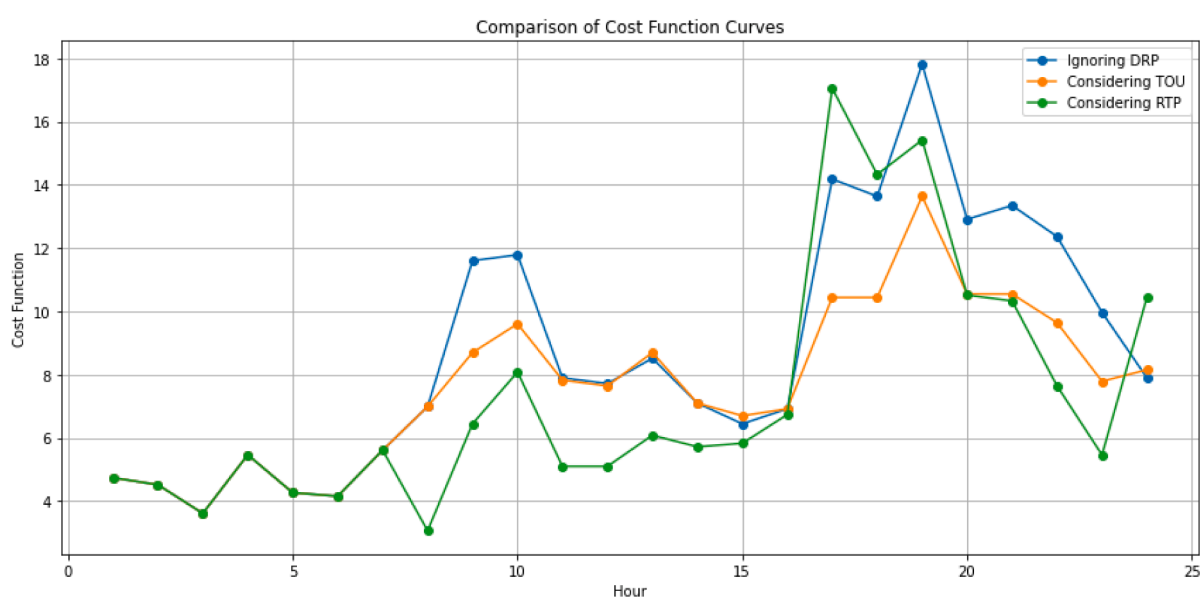

Fig. 5. Expense Function Assessment.

progress, a testament to human ingenuity, and a harbinger of a brighter, more energy-efficient tomorrow.

The synergy of advanced optimization techniques and the learning capabilities of recurrent neural networks (RNNs) enhances the resilience, efficiency, and adaptability of microgrid operations in stochastic demand response scenarios. Advanced optimization techniques, such as the modified covariance matrix adaptation, address uncertainties by capturing and adapting to variations in demand patterns and external factors. The RNN adds intelligence by learning from historical data and capturing temporal dependencies, resulting in more accurate predictions. This adaptive learning equips the microgrid with the ability to respond to evolving demand patterns, improving overall reliability in scheduling optimization. In stochastic scenarios where variables exhibit unpredictable behavior, this hybrid approach allows for nuanced decision-making, providing a robust framework for managing uncertainties in real-world microgrid operations.

Conclusions

In conclusion, the exploration of Stochastic Demand Response Scheduling Optimization in a SMG, leveraging a Hybrid Modified Covariance Matrix Adaptation and Recurrent Neural Network (RNN) Technique, represents a significant stride towards enhancing the resilience, efficiency, and adaptability of MG operations. The synergy of advanced optimization techniques, such as the modified covariance matrix adaptation, with the learning capabilities of recurrent neural networks, provides a robust framework for addressing the inherent uncertainties in demand response scenarios. This research has illuminated the potential for improving scheduling strategies in smart MGs by incorporating stochastic elements and harnessing the power of machine learning. The hybrid approach offers a nuanced understanding of complex, dynamic energy systems, allowing for adaptive decision-making in response to varying demand patterns and external factors. The integration of stochastic elements acknowledges the inherent unpredictability in demand response scenarios, where variables such as energy consumption and generation exhibit stochastic behavior. The modified covariance matrix adaptation provides a mechanism to capture and adapt to these uncertainties, ensuring that scheduling decisions are not only optimized for current conditions but also resilient to unforeseen variations. The utilization of a Recurrent Neural Network adds a layer of intelligence to the optimization process. By learning from historical data and capturing temporal dependencies, the RNN contributes to more accurate predictions and, consequently, improved scheduling results. This adaptive learning capability equips the MG with the ability to respond to evolving demand patterns and enhances the overall reliability of the scheduling optimization process. As we look to the future of smart MGs, characterized by increasing complexity and dynamic interactions, the findings from this study underscore the importance of embracing stochastic optimization and machine learning techniques. These tools not only address the challenges posed by uncertainties but also pave the way for more sophisticated, data-driven decision-making in the realm of demand response scheduling. In essence, the Hybrid Modified Covariance Matrix Adaptation and RNN Technique presented in this research not only contributes to the academic discourse on SMG optimization but also holds practical implications for real-world applications. As we strive for sustainable, resilient, and intelligent energy systems, the insights gleaned from this study provide a valuable foundation for shaping the future of demand response scheduling in smart MGs. In summary, integrating Digital Twin technology into Non-Intrusive Load Monitoring (NILM) and smart metering devices for Demand Response (DR) represents a pivotal advancement in the energy sector. The use of Digital Twins enhances operational efficiency, offering real-time monitoring, analysis, and predictive capabilities for smart meters. This is particularly valuable in optimizing energy distribution strategies and fine-tuning Demand Response initiatives for grid stability and user satisfaction. The combination of Digital Twins and NILM provides a sophisticated platform for accurate energy disaggregation and personalized insights for consumers. Additionally, Digital Twins play a crucial role in fortifying cybersecurity measures, ensuring the integrity and resilience of the grid. Overall, this integration contributes to a smarter, more adaptive energy grid, laying the groundwork for intelligent and responsive energy management in the future.

CRediT authorship contribution statement

Xiaoli Zhao: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Yiyang Zhang:** Data curation, Visualization, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A

In above formulations, S is the user award. To maximize the user's award:

$$\frac{\partial S}{\partial P_{I,new}(i)} = 0 \tag{6}$$

Applying Eq.5 and Eq.6 yields the following result:

$$\frac{\partial B(P_{I,new}(i))}{\partial P_{I,new}(i)} - C_{I,new}(i) = 0 \tag{7}$$

$$\frac{\partial B(P_{I,new}(i))}{\partial P_{I,new}(i)} = C_{I,new}(i) \tag{8}$$

Here B is user's income. Follwoing Taylor Series expansion of $B(P_{l,new}(i))$ [15]:

$$B(P_{I,new}(i)) = B(P_{I0}(i)) + C_{I0}(i) \times \left[P_{I,new}(i) - P_{I0}(i)\right] \times \left\{1 + \frac{P_{I,new}(i) - P_{I0}(i)}{2E(i,i)P_{I0}(i)}\right\}, i \in T \tag{9}$$

E(i,i) displays self-electricity. When Eq.8 and Eq.9 are mixed:

$$C_{I,new}(i) = C_{I0}(i) \times \left\{1 + \frac{P_{I,new}(i) - P_{I0}(i)}{2E(i,i)P_{I0}(i)}\right\}, i \in T \tag{10}$$

Eq.11 can be deployed to get the single period framework for demand requests:

$$P_{I,new}(i) = P_{I0}(i) \times \left\{1 + E(i,i)\frac{C_{I,new}(i) - C_{I0}(i)}{C_{I0}(i)}\right\}, i \in T \tag{11}$$

Along with the second framework, loads are able to shift from a cycle to the another one, i.e., from high point to low periods. Such kind of framework is recognized as multi-sensitivity:

$$E(i,j) = \frac{C_{I0}(i)}{P_{I0}(i)} \times \frac{\partial P_{I,new}(i)}{\partial C_{I,new}(j)} \forall i \neq j, \forall i,j \in T \tag{12}$$

Since in (12) the load swings to expense swings are static, the DR to expense change is showed similar to a linear equation [16]:

$$P_{I,new}(i) = P_{I0}(i) + \sum_{j\, i \neq j} E(i,i)\frac{P_{I0}(i)}{C_{I0}(j)}\left[C_{I,new}(j) - C_{I0}(j)\right], i \in T \tag{13}$$

In the concluding framework, the solo period and multi-period frameworks are mixed to reach the time-based DRP:

$$P_{I,new}(i) = P_{I0}(i) \times \left\{1 + E(i,i)\frac{C_{I,new}(i) - C_{I0}(i)}{C_{I0}(i)}\right\} + P_{I0}(i) \times \sum_{j\, i \neq j} 1 + E(i,j)\frac{C_{I,new}(j) - C_{I0}(j)}{C_{I0}(j)}, \forall i \in T \tag{14}$$

The sole expenses for SOLAR and WT generation agents are associated with conservation and operation.

$$C^t_{OM,WT} = \lambda_{OM,WT} \times P^t_{g,WT} \forall t \in T \tag{15}$$

$$C^t_{OM,PV} = \lambda_{OM,PV} \times P^t_{g,PV} \forall t \in T \tag{16}$$

Here λ is expense coefficient. Fuel expenses for the fuel cell unit (FU) and micro turbine unit (MU) are calculated as:

$$C^t_{g,MT} = \lambda_{g,MT} \times \frac{P^t_{g,WT}}{\eta^t_{MT}} \forall t \in T \tag{17}$$

$$C^t_{g,FC} = \lambda_{g,FC} \times \frac{P^t_{g,FC}}{\eta^t_{FC}} \forall t \in T \tag{18}$$

$$\lambda_{g,MT} = \lambda_{g,FC} = \frac{C_{nl}}{L} \tag{19}$$

Where L and C_{nl} represent the low heating value of natural gas and the associated expense, respectively. In contrast to MU, FU experiences a decline in performance as generation levels increase. The conservation expenses for FU and MU over time intervals are suggested as follows:

$$C^t_{OM,MT} = \lambda_{OM,MT} \times P^t_{g,MT} \forall t \in T \tag{20}$$

$$C^t_{OM,FC} = \lambda_{OM,FC} \times P^t_{g,FC} \forall t \in T \tag{21}$$

$$u^t_{g,MT} \bullet P^t_{min,MT} \leq P^t_{g,MT} \leq u^t_{g,MT} \bullet P^t_{max,MT} \forall t \in T \tag{22}$$

$$u^t_{g,FC} \bullet P^t_{min,FC} \leq P^t_{g,FC} \leq u^t_{g,Fc} \bullet P^t_{max,FC} \forall t \in T \tag{23}$$

U is assurance status of producers, and $P^t_{g,MT}$, $P^t_{g,FC}$ are MU and FU result power. $P^t_{min,MT}$ and $P^t_{max,MT}$ are low and high result power of MU. $P^t_{min,FC}$ and $P^t_{max,FC}$ are low and high result power of FU. Apart from the maximum and minimum power limits, remaining MUs and FUs in any micro-grid have been limited inside their incline rate limitations:

$$P^t_{g,MT} - P^{t-1}_{g,MT} \leq UR^t_{g,MT}, if P^t_{g,MT} > P^{t-1}_{g,MT} \forall t \in T \tag{24}$$

$$P^t_{g,FC} - P^{t-1}_{g,FC} \leq UR^t_{g,FC}, if P^t_{g,FC} > P^{t-1}_{g,FC} \forall t \in T \tag{25}$$

$$P^{t-1}_{g,MT} - P^t_{g,MT} \leq DR^t_{g,MT}, if P^t_{g,MT} < P^{t-1}_{g,MT} \forall t \in T \tag{26}$$

$$P^{t-1}_{g,FC} - P^t_{g,FC} \leq DR^t_{g,FC}, if P^t_{g,FC} < P^{t-1}_{g,FC} \forall t \in T \tag{27}$$

UR/DR display incline up / down rate. $UR^t_{g,MT}$ and $UR^t_{g,FC}$ are incline up rate of MU and FU. $DR^t_{g,MT}$ and $DR^t_{g,FC}$ are incline down rate of MU and FU.

CHP effectiveness improves the presentation of MU parts and lessens their fuel expenses. Below is the energy expense for MU parts with CHP productivity:

$$C_{g,CHP} = C_{g,MT} \times \left(1 - \frac{\epsilon_{rec}\left(\eta^t_{CHP} - \eta^t_e\right)}{\eta_b}\right), \forall t \in T \tag{28}$$

Where ϵ_{rec} is heat recovery factor. C is energy expense in different situations. $C_{g,CHP}$, $C_{g,MT}$ are energy expense of CHP and MU in different situations. η^t_e and η_b are MU's electrical Efficiency at time t and boiler Efficiency. η is effectiveness of production pieces, and also η_{CHP} is effectiveness of CHP unit.

The charge level of a battery at hour t relies on its State of Charge (SOC) at the previous time step, in addition to the magnitude of energy filled and discharged.

$$0 \le P^t_{BAT,CH,m} \le P_{BAT,CAP,m} \times \left(1 - SOC^{t-1}_{BAT,m}\right) \times \frac{1}{1 - P^{loss}_{BAT,CH,m}} \forall t \in T, \forall m \in MG \tag{29}$$

$$0 \le P^t_{BAT,DCH,m} \le P_{BAT,CAP,m} \times SOC^{t-1}_{BAT,m} \times \left(1 - P^{loss}_{BAT,CH,m}\right), \forall t \in T, \forall m \in MG \tag{30}$$

$$SOC^t_{BAT,m} = SOC^{t-1}_{BAT,m} - \frac{1}{P_{BAT,CAP,m}} \times \left(\frac{1}{1 - P^{loss}_{BAT,DCH,m}} \times P^t_{BAT,DCH,m} - \left(1 - P^{loss}_{BAT,CH,m}\right) \times P^t_{BAT,CH,m}\right) \forall t \in T, \forall m \in MG \tag{31}$$

$$SOC_{min,BAT,m} \le SOC^t_{BAT,m} \le SOC_{max,BAT,m} \forall t \in T, \forall m \in MG \tag{32}$$

Where $P^{loss}_{BAT,CH,m}$ and $P_{BAT,CAP,m}$ are power loss of battery and storage capacity.

Consequently, the key principles in exchange relise on financial difficulties. This allows the (OF) to reach its most ideal worth for micro-grids. To trade energy, the boundaries below must be applied:

$$if P^t_{g,m} - P^t_l > 0 \rightarrow P^t_{pur,m} = 0, P^t_{sell,m} > 0, \forall t \in T, \forall m \in MG \tag{33}$$

$$if P^t_{g,m} - P^t_l < 0 \rightarrow P^t_{pur,m} > 0, P^t_{sell,m} = 0, \forall t \in T, \forall m \in MG \tag{34}$$

Where $P^t_{pur,m}$ and $P^t_{sell,m}$ are expense of traded power from micro-grid-m and traded power to micro-grid-m.

In supply lines, energy blocking is a critical problem. The power exchange boundary is the following:

$$0 \le P^t_{tran,m-n} \le P^t_{max,tran,m-n}, \forall t \in T, \forall \{m,n\} \in MG, m \ne n \tag{35}$$

Where $P^t_{tran,m}$ is the transitive power value of micro-grid-m. Underneath is the purchased power expense in micro-grid:

$$C^t_{pur,m} = \sum_n C^t_{pur,mn} \times P^t_{pur,mn} \quad \forall t \in T, \forall \{m,n\} \in \{MG, NW, BAT\}, m \ne n \tag{36}$$

Where $C^t_{pur,mn}$ is purchased power expense via micro-grid-m from micro-grid-n (\$/h). With the equation above, any MICRO-GRID trades power from 4 resources in line with its demand. The MICRO-GRID1 buys its power either from MICRO-GRID2/MICRO-GRID3, or from the battery packages, or from the key system relied on financial benefits. Any MICRO-GRID occupies in the market relied on its overall profits. MICRO-GRID1 at time t is showed by the equation below for allowance of Equation (36):

$$C^t_{pur,m} = C^t_{pur,12} \times P^t_{pur,12} + C^t_{pur,13} \times P^t_{pur,13} + C^t_{pur,1BAT} \times P^t_{pur,1BAT} + C^t_{pur,1NW} \times P^t_{pur,1NW} \forall t \in T \tag{37}$$

For traded energy expenses, the equal relations as for purchased power are used as below:

$$C^t_{sell,m} = \sum_n C^t_{sell,mn} \times P^t_{sell,mn} \forall t \in T, \forall \{m,n\} \in \{MG, NW, BAT\}, m \ne n \tag{38}$$

Where $C^t_{sell,mn}$ is traded energy expense via micro-grid-m to micro-grid-n (\$/h). As an effect of the variance among the expense of purchased and traded energy, the power transitions expense in any micro grid is as below:

$$C^t_{tran,m} = C^t_{pur,m} - C^t_{sell,m} \forall t \in T, \forall m \in MG \tag{39}$$

The following equations is employed to optimize the problem for each micro grid:

$$MinOF_m = \sum_t \left(Cost^t_{op,m} + Cost^t_{em,m}\right), \forall t \in T, \forall m \in \{MG1, MG2, MG3\} \tag{40}$$

$$Cost^t_{op,m} = C^t_{g,m} + C^t_{tran,m} + C^t_{OM,m}, \forall t \in T, \forall m \in \{MG1, MG2, MG3\} \tag{41}$$

$$Cost^{t}_{op,m} = \sum_{k} \gamma_k \times \left(\sum_{u} \rho_{uk} \times P^{t}_{g,u} \right), \forall t \in T, \forall k \in \{CO_2, SO_2, NO_x\}, \forall m \in \{MG1, MG2, MG3\}, \forall u \in \{WT, PV, FC, MT, CHP\} \tag{42}$$

Where OF defines the OF. $Cost^{t}_{op}$ and $Cost^{t}_{em}$ are procedure expense (OC) [$/h] and emission expense [$/h]. ρ and γ are discharge cause of emissions and expense constant of diverse emissions.

Within any specified time period, the total power produced and transmitted power, along with the discharge/charge status of batteries, must collectively satisfy the predicted power demand of each MG:

$$P^{t}_{g,m} = P^{t}_{g,m} + P^{t}_{tran,m} + P^{t}_{CH,m} + P^{t}_{DCH,m}, \forall t \in T, \forall m \in \{MG1, MG2, MG3\} \tag{43}$$

Appendix B

Next Hansen and Ostermeier, in (μ_1, λ)-CMA-ES the λ offspring of generation $g+1$ are computed by

$$x_k^{(g+1)} = \langle x \rangle_{\mu}^{(g)} + \sigma^{(g)} \underbrace{B^{(g)} D^{(g)} z_k^{(g+1)}}_{\sim N\left(0, C^{(g)}\right)}, k = 1, \ldots, \lambda, \tag{54}$$

where

$$\langle x \rangle_{\mu}^{(g)} = \frac{1}{\mu} \sum_{i \in I^{(g)}_{\sim -1}} x_i^{(g)} \tag{55}$$

Denotes the focal point of mass for the chosen individuals within the production g, and $I^{(g)}_{sel}$ comprises the array of indices for the chosen individuals within the production. g with $\left| I^{(g)}_{\text{sel}} \right| = \mu \cdot \sigma^{(g)}$ is the global step size. Also the center of mass is as below:

$$\langle z \rangle_{j}^{(g+1)} = \frac{1}{\mu} \sum_{i \in I^{(g+1)}_{a=1}} z_i^{(g+1)}. \tag{56}$$

The covariance matrix $C^{(g)}$ of the random vectors $B^{(g)} D^{(g)} z_k^{(g+1)}$ is a symmetrical positive $n \times n$-matrix. The columns of the orthogonal matrix $B^{(a)}$ represent normalized eigenvectors of the covariance matrix. $D^{(g)}$ is a diagonal matrix whose elements are the square roots of the eigenvalues of $C^{(p)}$. Hence, the relation of $B^{(p)}$ and $D^{(p)}$ to $C^{(p)}$ can be expressed by

$$C^{(g)} = B^{(g)} D^{(g)} \left(B^{(g)} D^{(g)} \right)^{\mathrm{T}} \text{ and } C^{(g)} b_i^{(p)} = \left(d_{ii}^{(g)} \right)^2 \cdot b_i^{(p)} \tag{57}$$

where $b_i^{(g)}$ represents the i-th column of $B^{(g)}$ and $\left\| b_i^{(g)} \right\| = 1$ and $d_{ii}^{(p)}$ are the diagonal elements of $D^{(a)}$. Surfaces of equal probability density of the random vectors $B^{(g)} D^{(g)} z_k^{(g+1)} \sim \mathcal{N}\left(0, C^{(g)}\right)$ spheroids or hyperellipsoids are geometric shapes defined by their main axes aligning with the eigenvectors of a covariance matrix. The squared magnitudes of these axes precisely match the eigenvalues of the covariance matrix. Subsequent section delineates the 2 adjustment mechanisms within the CMA-ES: (i) the modification of matrix covariance $C^{(g)}$ as well as (ii) the adjustment of size global step $\sigma^{(a)}$.

Initially, the progression path $p_c^{(p+1)}$ is computed by

$$p_c^{(g+1)} = (1 - c_c) \cdot p_c^{(g)} + \sqrt{c_c \cdot (2 - c_c)} \cdot \underbrace{\frac{\sqrt{\mu}}{\sigma^{(g)}} \left(\langle x \rangle_{\mu}^{(g+1)} - \langle x \rangle_{\mu}^{(g)} \right)}_{= \sqrt{\mu} B^{(g)} D^{(g)} \langle z \rangle_{\mu}^{(g+1)}} \tag{58}$$

and has been utilized to create matrix covariance of generation $g + 1$

$$C^{(g+1)} = (1 - c_{\text{cov}}) \cdot C^{(g)} + c_{\text{cov}} \cdot p_c^{(g+1)} \left(p_c^{(g+1)} \right)^{\mathrm{T}}. \tag{59}$$

Matrix C undergoes a modification by adding a symmetric matrix of rank one, represented as the right summand in Equation (59). Please note that when utilizing c_c is equal to 1 at the Equation (58), the evolution path simplifies to $\sqrt{\mu} BD \langle z \rangle_{\mu}$, representing mutation step mean of latest group reproduced by $\sqrt{\mu}$. Opting for $c_c < 1$ typically accelerates and enhances adaptation as it leverages the correlation between successive steps. This research has even originate it helpful to select $c_c \propto \frac{1}{n}$ due to this configuration attains the linear grading of the plan on f_{cigar} .

Next, in order to adjust step size global σ, the path of evolution $p_{\sigma}^{(g+1)}$ has been calculated similarly to path of evolution $p^{(g+1)}$. Distinction lies in fact that $p_{\sigma}^{(g+1)}$ is not scaled by $D^{(g)}$, resulting in

$$\boldsymbol{p}_{\sigma}^{(g+1)} = (1-c_{\sigma})\cdot\boldsymbol{p}_{\sigma}^{(g)} + \sqrt{c_{\sigma}\cdot(2-c_{\sigma})}\cdot\underbrace{\sqrt{\mu}\boldsymbol{B}^{(g)}\langle z\rangle_{\mu}^{(g+1)}}.$$
$$= \boldsymbol{B}^{(g)}\left(\boldsymbol{D}^{(g)}\right)^{-1}\left(\boldsymbol{B}^{(g)}\right)^{-1}\frac{\sqrt{\mu}}{\sigma^{(g)}}\left(\langle \boldsymbol{x}\rangle_{\mu}^{(g+1)} - \langle \boldsymbol{x}\rangle_{\mu}^{(g)}\right) \tag{60}$$

$$\sigma^{(g+1)} = \sigma^{(g)}\cdot\exp\left(\frac{1}{d_{\sigma}}\frac{p_{\sigma}^{(g+1)} - \overline{\chi}_n}{\overline{\chi}_n}\right) \tag{61}$$

where $\overline{\chi}_n = \mathrm{E}[\|\mathcal{N}(0,I)\|]$ is the expected length of a $(0,I)$-normally distributed random vector and $d_{\sigma} > 1$ is the damping parameter. $\overline{\chi}_n$ is approximated by $\overline{\chi}_n \approx \sqrt{n}\left(1 - \frac{1}{4n} + \frac{1}{11n^2}\right)$ (Ostermeier, 1997):

$$c_{\mathrm{c}} = \frac{4}{n+4},\ c_{\mathrm{cor}} = \frac{2}{\left(n+\sqrt{2}\right)^2},\ c_{\sigma} = \frac{4}{n+4},\ d_{\sigma} = c_{\sigma}^{-1} + 1 \tag{62}$$

Initial values are $p^{(\mathrm{o})} = 0, p_{\sigma}^{(\mathrm{o})} = 0$ and the initial covariance matrix $C^{(0)}$ is the identity matrix $\boldsymbol{I}$. The problem dependent initial values for $\langle x\rangle_j^{(\phi)}$ and $\sigma^{(0)}$ are given in Section 5.

Adapting the (nˆ2 + n)/2 elements of the covariance matrix in CMA-ES involves a number of function evaluations that scales between O(n) and O (nˆ2) (Hansen and Ostermeier, 2001), depending on the function topology. In basic CMA-ES, the covariance matrix is updated in each group by adding the (weighted) outer product of the evolution path, which is a symmetrical n × n matrix of rank one.

The concept behind altering the algorithm is to adjust the covariance matrix by making better use of the information inherent in larger populations. Rather than updating the covariance matrix solely with rank one information, we incorporate higher-rank information. This is accomplished by revising Equation (59), governing the covariance matrix's change (i.e., the alteration of the mutation distribution shape), and adjusting the strategy parameter c_{cor}. The adaptation of the global step size (as per Equations (60) and (61)) remains unaltered.

The adjustment is gotten by adding to (63) to (6):

$$Z^{(g+1)} = \frac{1}{\mu}\sum\nolimits_{i\in I_{m=1}^{(g+1)}} B^{(g)}D^{(g)}z_i^{(g+1)}\left(B^{(g)}D^{(g)}z_i^{(g+1)}\right)^{\mathrm{T}}$$
$$= B^{(g)}D^{(g)}\left(\frac{1}{\mu}\sum\nolimits_{i\in I_{m=1}^{(g+3)}} z_i^{(g+1)}\left(z_i^{(g+1)}\right)^{\mathrm{T}}\right)\left(B^{(g)}D^{(g)}\right)^{\mathrm{T}} \tag{63}$$

that is a symmetrical $n \times n$ matrix with $\mathrm{rankmin}(\mu, n)$.

Equation (59) is substituted by the new adaptation of the covariance matrix:

$$C^{(g+1)} = (1-c_{\mathrm{cor}})\cdot C^{(g)} + c_{\mathrm{cor}}\left(\alpha_{\mathrm{cow}}\cdot p_{\mathrm{c}}^{(g+1)}\left(p_{\mathrm{c}}^{(g+1)}\right)^{\mathrm{T}} + (1-\alpha_{\mathrm{cor}})\cdot Z^{(g+1)}\right) \tag{64}$$

where $0 \le \alpha_{\mathrm{cor}} \le 1$. Hence

$$c_{\mathrm{cov}} = \alpha_{\mathrm{cov}}\frac{2}{\left(n+\sqrt{2}\right)^2} + (1-\alpha_{\mathrm{cov}})\min\left(1, \frac{2\mu-1}{(n+2)^2+\psi}\right). \tag{65}$$

Note, that for $\alpha_{\mathrm{cor}} = 1$ unique c_{cov} value is recovered.

Under those chosen expectations, the symmetric matrix

$$\sum_{i\in I_{a+1}} z_i(z_i)^{\mathrm{T}} = \sum_{i\in I_{a+1}}\begin{pmatrix} z_{i1}^2 & z_{i1}z_{i2} & \cdots & z_{i1}z_{in} \\ z_{i2}z_{i1} & z_{i2}^2 & \cdots & z_{i2}z_{in} \\ \vdots & \vdots & \ddots & \vdots \\ z_{in}z_{i1} & z_{in}z_{i2} & \cdots & z_{in}^2 \end{pmatrix} \tag{66}$$

Considering $E\left[\sum_{i\in I_{a+1}} \boldsymbol{z}_{ij}^2\right] = \mu, j = 1, \cdots n$, we have that

$$\mathrm{E}\left[Z^{(g+1}\right] = \frac{1}{\mu}B^{(g)}D^{(g)}\mu I\left(B^{(g)}D^{(g)}\right)^{\mathrm{T}} = C^{(g)} \tag{67}$$

under the given chosen part frameworks. Equation (14) is aim to select coefficient $\frac{1}{\mu}$ at the Equation (10). Considering $\mathrm{E}\left[p_c^{(g+1)}\left(p_c^{(g+1)}\right)^{\mathrm{T}}\right] = C^{(g)}$ (Hansen 1998) and $\mathrm{E}\left[Z^{(p+1)}\right] = C^{(p)}$, we conclude from Equation (11) that $\mathrm{E}\left[C^{(p+1)}\right] = C^{(g)}$. This is the motive for choosing the constants $(1-\alpha_{\mathrm{cor}})$ in combination with α_{cor} at the Equation (64). The total summation should be equal to 1.

References

[1] Aguila-Leon J, Vargas-Salgado C, Díaz-Bello D. Energy Management Framework for a Standalone Hybrid Microgrid through a Particle Swarm Optimization and Artificial Neural Networks Approach. Energy Conversion and Management; 2022.

[2] Akinyele DO, Rayudu RK. Sustainable Energy Technologies and Assessments. Sustainable Energy Technol Assess 2014;8:74–91.

[3] Dabbaghjamanesh M, Kavousi-Fard A, Zhang J. Stochastic modeling and integration of plug-in hybrid electric vehicles in reconfigurable microgrids with deep learning-based forecasting. IEEE Trans Intell Transp Syst 2020;22(7): 4394–403.

[4] Nagasri LS, Marimuthu D. October 3). "Review on advanced control techniques for.". Energy Rep 2023.

[5] Khalil, Muhammad Ibrahim, N. Z. Jhanjhi, Mamoona Humayun, SivaKumar Sivanesan, Mehedi Masud, and M. Shamim Hossain. "Hybrid smart grid with sustainable energy efficient resources for smart cities." sustainable energy technologies and assessments 46 (2021): 101211.

[6] Khalid, R., Javaid, N. (2020, June 4). "A survey on hyperparameters optimization algorithms of forecasting frameworks in smart grid." Sustainable Cities and Society.

[7] Debouza M, Al-Durra A, Zeineldin HH. Survey on with flexible boundaries: Strategies, applications, and future trends. Electric Power Systems Research; 2022.

[8] Huang W, Dai J, Xiong L. Towards a sustainable energy future: Factors affecting solar-hydrogen energy production in China. Sustainable Energy Technol Assess 2022;52:102059.

[9] Zheng D, Zhang W, Yue J. Chapter 4: Short-term renewable generation and load forecasting in. Microgrid Protection and Control; 2021.

[10] Alanis, A. Y., Arana-Daniel, N., López-Franco, C. (2018). "Chapter Six: Particle Swarm Optimization to Improve Neural Identifiers for Discrete-time Unknown Nonlinear Systems." Bio-inspired Algorithms for Engineering.

[11] Alsirhani A, Alshahrani MM, Salem M. May 23). "A novel approach to predicting the stability of the smart grid utilizing MLP-ELM technique.". Alex Eng J 2023.

[12] Mousaei A, Gheisarnejad S, Khooban MH. September 4). "Challenges and opportunities of FACTS devices interacting with electric vehicles in distribution networks: A technological review.". J Storage Mater 2023.

[13] Himeur, Y., Elnour, M., Amira, A. (2022, July 19). "Next-generation energy systems for sustainable smart cities: Roles of transfer learning." Sustainable Cities and Society.

[14] Adnan N, Shahrina MN. A comprehensive approach: Diffusion of environment-friendly energy technologies in residential photovoltaic markets. Sustainable Energy Technol Assess 2021;46:101289.

[15] Gálvez J, Cuevas E, Pérez-Cisneros M. May 1). "A reactive framework based on neighborhood consensus for continuous optimization.". Expert Systems with Applications. 2019.

[16] Ghorvei M, Kavianpour M, Ramezani A. October 30). "Spatial graph convolutional neural network via structured subdomain adaptation and domain adversarial learning for bearing fault diagnosis.". Neurocomputing 2022.

[17] Crisostomi, E., Gallicchio, C., Tucci, M. (2015, December 25). "Prediction of the Italian electricity expense for smart grid applications." Neurocomputing.

[18] Bose A, Hsu C-H, Abimannan S. September 9). "Forecasting stock expense by hybrid framework of cascading Multivariate Adaptive Regression Splines and Deep Neural Network.". Comput Electr Eng 2021.

[19] Martínez-López, Y., Rodríguez-González, A. Y., Lezama, F. (2021, March 20). "Cellular estimation of distribution algorithm designed to solve the energy resource management problem under uncertainty." Engineering Applications of Artificial Intelligence.

[20] Gong, X., Wang, X., Cao, B. (2023, August 23). "On data-driven frameworking and control in modern power grids stability: Survey and perspective." Applied Energy.

[21] Singh V, Moger T, Jena D. October 25). "Uncertainty handling techniques in power systems: A critical review.". Electr Pow Syst Res 2021.

[22] Mahmoud, M. S., Xia, Y. (2019, February 15). "Chapter 6: Examples." Networked Control Systems.

[23] Li, H. (2016). "Chapter 7: Communication network operation for CPSs." Communications for Control in Cyber Physical Systems.

[24] Fei, C., Shen, J. (2023, September). "Machine learning for securing Cyber–Physical Systems under cyber attacks: A survey." Franklin Open.

[25] Xizheng Zhang, Zeyu Wang, Zhangyu Lu. "Multi-objective load dispatch for microgrid with electric vehicles using modified gravitational search and particle swarm optimization algorithm." Applied Energy11 October 2021.